K-Means Clustering — Proof of Equation (12.18) and Monotonic Descent

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The notation and equation numbering in this solution follow James et al. (2023). Let $\mathbf{C_k}$ denote the set of observations assigned to cluster $\mathbf{k}$, let $\mathbf{n_k} = |\mathbf{C_k}|$, and define the mean of feature $\mathbf{j}$ within cluster $\mathbf{k}$ as

$$\bar{\mathbf{x}}_{\mathbf{kj}} = \frac{\mathbf{1}}{\mathbf{n_k}} \sum_{\mathbf{i} \in \mathbf{C_k}} \mathbf{x_{ij}}.$$

(a) Proof of Equation (12.18)

Equation (12.18) states that

$$\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} \left(x_{ij} - x_{i'j}\right)^2 = 2 \sum_{i \in C_k} \sum_{j=1}^{p} \left(x_{ij} - \bar{x}_{kj}\right)^2.$$

Consider one feature j within a fixed cluster C_k. Expanding the squared difference gives

$$\sum_{i,i' \in C_k} \left(x_{ij} - x_{i'j}\right)^2 = \sum_{i,i' \in C_k} \left(x_{ij}^2 + x_{i'j}^2 - 2x_{ij}x_{i'j}\right).$$

Because each value x_{ij}^2 appears n_k times as the other index varies,

$$\sum_{i,i' \in C_k} x_{ij}^2 = n_k \sum_{i \in C_k} x_{ij}^2$$

and, by the same reasoning,

$$\sum_{i,i' \in C_k} x_{i'j}^2 = n_k \sum_{i \in C_k} x_{ij}^2.$$

The cross-product term satisfies

$$\sum_{i,i' \in C_k} x_{ij}\, x_{i'j} = \left(\sum_{i \in C_k} x_{ij}\right)^2.$$

Combining these three results yields

$$\sum_{i,i'\in C_k}\left(x_{ij}-x_{i'j}\right)^2 = 2n_k\sum_{i\in C_k}x_{ij}^2 - 2\left(\sum_{i\in C_k}x_{ij}\right)^2.$$

After dividing by n_k,

$$\frac{1}{n_k}\sum_{i,i'\in C_k}\left(x_{ij}-x_{i'j}\right)^2 = 2\sum_{i\in C_k}x_{ij}^2 - \frac{2}{n_k}\left(\sum_{i\in C_k}x_{ij}\right)^2.$$

Since $\sum_{i\in C_k}x_{ij} = n_k\bar{x}_{kj}$, the preceding expression becomes

$$\frac{1}{n_k}\sum_{i,i'\in C_k}\left(x_{ij}-x_{i'j}\right)^2 = 2\left(\sum_{i\in C_k}x_{ij}^2 - n_k\bar{x}_{kj}^2\right).$$

Now expand the squared deviations from the cluster mean:

$$\sum_{i\in C_k}\left(x_{ij}-\bar{x}_{kj}\right)^2 = \sum_{i\in C_k}\left(x_{ij}^2 - 2x_{ij}\bar{x}_{kj} + \bar{x}_{kj}^2\right).$$

Collecting the terms gives

$$\sum_{i\in C_k}x_{ij}^2 - 2\bar{x}_{kj}\sum_{i\in C_k}x_{ij} + n_k\bar{x}_{kj}^2.$$

Using $\sum_{i\in C_k}x_{ij} = n_k\bar{x}_{kj}$ once more, this expression simplifies to

$$\sum_{i\in C_k}\left(x_{ij}-\bar{x}_{kj}\right)^2 = \sum_{i\in C_k}x_{ij}^2 - n_k\bar{x}_{kj}^2.$$

Therefore,

$$\frac{1}{n_k}\sum_{i,i'\in C_k}\left(x_{ij}-x_{i'j}\right)^2 = 2\sum_{i\in C_k}\left(x_{ij}-\bar{x}_{kj}\right)^2.$$

Summing this identity over all p features proves Equation (12.18):

$$\boxed{\frac{1}{|C_k|}\sum_{i,i'\in C_k}\sum_{j=1}^{p}\left(x_{ij}-x_{i'j}\right)^2 = 2\sum_{i\in C_k}\sum_{j=1}^{p}\left(x_{ij}-\bar{x}_{kj}\right)^2}.$$

(b) Why K-Means Decreases the Objective at Each Iteration

The objective in Equation (12.17) is

$$\sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} \left(x_{ij} - x_{i'j}\right)^2.$$

Applying Equation (12.18) to every cluster rewrites this objective as

$$2\sum_{k=1}^{K} \sum_{i \in C_k} \sum_{j=1}^{p} \left(x_{ij} - \bar{x}_{kj}\right)^2 = 2\sum_{k=1}^{K} \sum_{i \in C_k} \parallel x_i - \bar{x}_k \parallel^2.$$

The factor of $\mathbf{2}$ is positive and constant, so minimizing Equation (12.17) is equivalent to minimizing the within-cluster sum of squares

$$W(C, \mu) = \sum_{k=1}^{K} \sum_{i \in C_k} \parallel x_i - \mu_k \parallel^2.$$

Algorithm 12.2 alternates between two steps, and each step minimizes $\mathbf{W(C, \mu)}$ with respect to one set of quantities while holding the other fixed.

Assignment Step

With the centroids $\boldsymbol{\mu_1}, \ldots, \boldsymbol{\mu_K}$ held fixed, each observation is assigned to the cluster with the nearest centroid:

$$c_i = \underset{k \in \{1,\ldots,K\}}{\arg\min} \parallel x_i - \mu_k \parallel^2.$$

This choice minimizes the contribution of observation $\mathbf{i}$ to $\mathbf{W(C, \mu)}$. Because the minimization is performed separately for every observation, the new assignments cannot increase the total within-cluster sum of squares. The value decreases strictly whenever at least one reassigned observation moves to a strictly closer centroid.

Centroid-update step

With the assignments held fixed, Algorithm 12.2 replaces each centroid by the mean of the observations in that cluster. To see why the mean is optimal, consider any candidate centroid $\mathbf{m_k}$. The following decomposition holds:

$$\sum_{i \in C_k} \| x_i - m_k \|^2 = \sum_{i \in C_k} \| x_i - \bar{x}_k \|^2 + n_k \| m_k - \bar{x}_k \|^2.$$

The second term is nonnegative and equals zero only when $m_k = \bar{x}_k$. Hence,

$$\bar{x}_k = \arg\min_{m_k} \sum_{i \in C_k} \| x_i - m_k \|^2.$$

Replacing every centroid with its cluster mean therefore cannot increase $\mathbf{W(C, \mu)}$. The value decreases strictly whenever a previous centroid differs from the mean of its currently assigned observations.

Conclusion

The assignment step does not increase the within-cluster sum of squares, and the centroid-update step does not increase it either. Equation (12.18) establishes that the objective in Equation (12.17) is exactly twice this within-cluster sum of squares. Consequently, each complete K-means iteration produces

$$\text{Objective}^{(t+1)} \leq \text{Objective}^{(t)}.$$

Thus, the statement that K-means “decreases” the objective should be understood as monotonic non-increase: the objective decreases until the cluster assignments and centroids stabilize, after which it remains unchanged. Because only finitely many cluster assignments are possible, Lloyd’s algorithm terminates at a stable partition under a deterministic tie-breaking rule. The solution is generally a local minimum rather than a guaranteed global minimum, which is why multiple random initializations are commonly used.

References

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An introduction to statistical learning: With applications in Python*. Springer. https://doi.org/10.1007/978-3-031-38747-0